

Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning: A Performance Comparison of ANN, Random Forest, and SVR

Ishaan Ghosh

Department of Chemical Engineering
Malaviya National Institute of Technology Jaipur

April 13, 2026

1 Problem statement

Distillation is an energy intensive separation process that requires rigorous design, which must also be optimised using complex Thermodynamic correlations. Software like DWSIM provides the environment to create flowsheets to calculate results after providing inputs with varying complexities. These calculations are made using the Equations of state like the Peng-Robinson method. While this software is helpful, optimising the process often needs a large number of iterations to compare which becomes time consuming. Therefore, Python automation using DWSIM COM bridge can reduce the computational time and also help get results as accurate as those obtained using intense thermodynamic calculations.

This project's aim was to generate a large enough dataset which helps in creation of precise and accurate machine learning models. By comparing Artificial Neural Networks (ANN), Support Vector Regression (SVR) and Random Forest (RF) models, this study finds the optimal model that is capable of predicting the outputs accurately for a given range of inputs.

2 DWSIM Flowsheet and Model description

The flowsheet shown below, **Figure1** was modeled in DWSIM (v9.0.5), designed to separate a mixture of Benzene and Toluene using steady state thermodynamic calculations.

2.1 Thermodynamic Framework

To calculate our desired outputs at any given input, an Equation of State (EOS) is used. For instance, this model uses the Peng-Robinson EOS which is a cubic equation, whose roots help in identifying the exact molar volume vapour, liquid and a factor bridging the two phases. The Peng-Robinson equation also takes acentric factor (ω) into consideration which does not assume the molecules to be spherical in nature, instead takes into account the non sphericity of each molecule.

2.2 Column Configuration

The separation is achieved using a distillation column block configured with the following characteristics:

- **Condenser Type:** Total condenser, ensuring the distillate product is drawn entirely as a liquid.
- **Reboiler Type:** Partial reboiler, adding heat to the bottom of the column to generate vapor traffic.
- **Feed State:** The feed conditions in each iteration of calculation was made sure to keep it as subcooled or saturated liquid to ensure stable convergence.

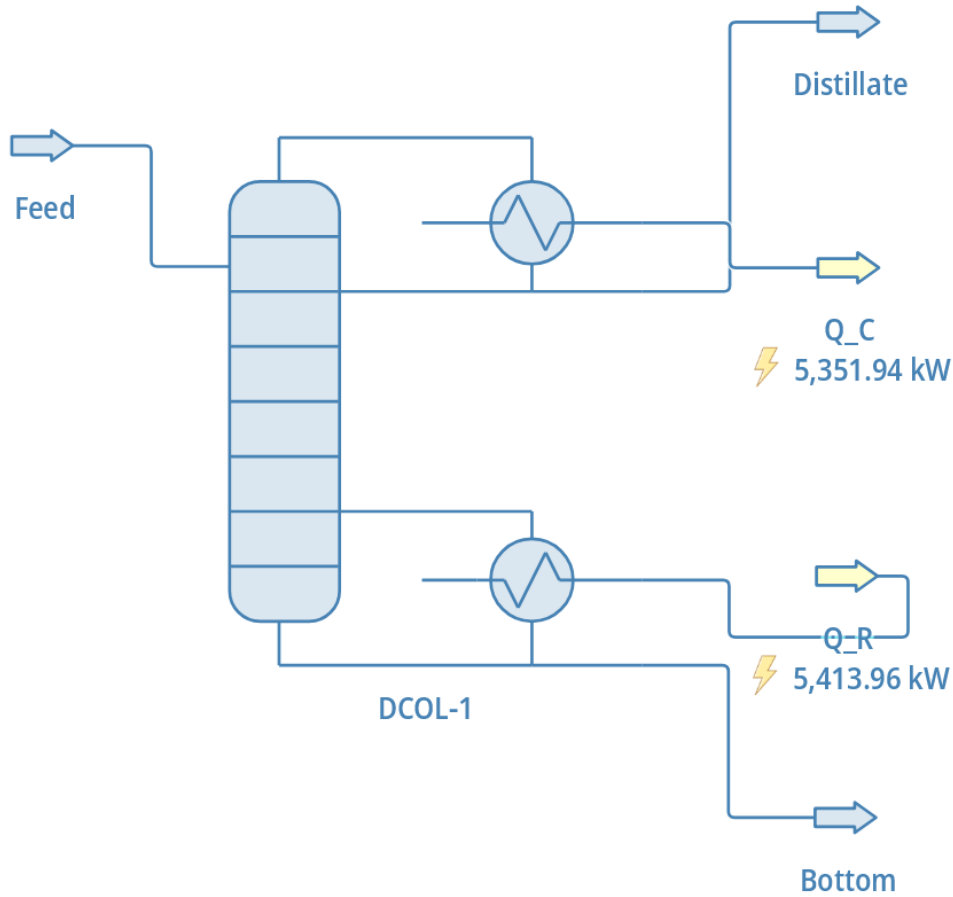


Figure 1: DWSIM flowsheet showing binary distillation of a mixture of Benzene and Toluene (Feed) into Distillate and Bottom product with the condenser and reboiler duties highlighted.

3 Input and Output Variables Used

Seven input parameters were taken to represent the operational degrees of freedom. These parameters represent the feed conditions and column geometry. Four output parameters were calculated using them representing the separation efficiency and energy requirements.

Input Parameters (Independent Variables):

- Feed Temperature ($Feed.T$)
- Feed Pressure ($Feed.P$)
- Liquid Feed Composition ($x_F(B)$, Benzene mole fraction)
- Total Number of Stages ($Stages$)

- Feed Input Stage ($Feed_Stage$)
- Reflux Ratio (RR)
- Bottom Withdrawal Rate (W)

Output Parameters (Dependent Variables):

- Condenser Duty (Q_{c_kW})
- Reboiler Duty (Q_{r_kW})
- Distillate Composition ($x_{D(B)}$, Benzene mole fraction)
- Bottoms Composition ($x_{W(T)}$, Toluene mole fraction)

4 Dataset Generation Strategy

Considering that changing the seven input variables per iteration done manually in DWSIM takes 2 minutes each, which is very much improbable since manual inputs need thought process behind each variable. Thus, $2\text{mins} \times 5000 \approx 7$ days of continuous work to get a dataset with 5000 rows.

Instead the data generation was automated using Python script integrated with DWSIM COM (Component Object Model) interface via `pythonnet` library. Each iteration took 2.7 seconds, which is about 3 hours and 50 minutes of data generation.

4.1 Sampling Methodology

To efficiently explore the 7-dimensional input space Latin Hypercube Sampling (LHS) was employed. A standard grid search with just 5 increments per variable would result in $5^7 = 78,125$ simulations, many of which would be computationally redundant. LHS divides the minimum to maximum range of each variable into n number of equal bins, in this case $n = 5000$. It then places points such that every single bin of every single variable has exactly one data point. This ensures that no two data points can match each other.

4.2 Automation Pipeline

The Python-DWSIM automation pipeline executed the following loop for each of the 5,000 sampled parameter sets:

1. **Injection:** Python generated the LHS numbers, providing meaningful range to each variable and pushed them into DWSIM. .
2. **Convergence:** The script commands DWSIM to iterate the variables by solving the flowsheet using the Peng-Robinson method.
3. **Extraction:** Upon successful convergence, Python reads the 4 dependent output variables (condenser/reboiler duties and product purities).
4. **Logging:** The input-output vector is appended to a pandas DataFrame and ultimately exported as a master CSV file (`DWSIM_ML_Dataset.csv`).

The input variables were provided calculated ranges to make sure each iteration was sure to converge and provide meaningful result.

5 Range of operating conditions considered

To ensure the machine learning models could predict actual non linear thermodynamic relationship rather than noise, the Latin Hypercube Sampling (LHS) was strictly bounded. The upper and lower limits for each independent variable were chosen to maintain convergence across all 5000 rows of data. One way of ensuring this was to keep the feed in the subcooled to saturated liquid region, avoiding any chances of divergence and hence preventing noise. The simulation ranges are detailed in Table 1.

Table 1: Range of operating conditions for parametric data generation

Variable	Unit	Minimum Bound	Maximum Bound
Feed Temperature	K	355.0	365.0
Feed Pressure	Pa	101,325	202,650
Feed Composition (Benzene)	Mole Fraction	0.40	0.60
Number of Trays	-	10	20
Feed Tray	-	2	18
Reflux Ratio	-	1.5	5.0
Bottoms Withdrawal Rate	kg/h	40.0	60.0

6 Data Preprocessing Steps

The raw 5000 row data underwent data preprocessing to ensure model stability and accuracy.

6.1 Data Cleaning

To ensure that the dataset had no individual values containing NaN (Not a Number) or zero values, the raw CSV was imported into Python via the **pandas** library, the mentioned numbers were programmed to be dropped.

6.2 Feature Scaling

The physical scales of the independent variables varied by several orders of magnitude (e.g., Feed Pressure at 10^5 Pa versus Feed Composition at 10^{-1} mole fraction). These ranges were standardised in order to prevent weighing the variables with a larger scale disproportionately.

This was done by Scikit-Learn’s **StandardScaler** which was applied to both the input features (X) and the target outputs (y). The scaler standardizes the data by removing the mean and scaling to unit variance using the Z -score formula:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Where μ is the mean of the training samples and σ is the standard deviation. The scaler states were saved as **Scaler_X.pkl** and **Scaler_Y.pkl** to ensure consistent transformation during future deployments.

7 Machine Learning Methods Implemented

To approximate the non linear thermodynamic relationship of the DWSIM calculations, three machine learning models were utilised. This was done to compare results and identify the optimal model that could replicate the correlations better.

7.1 Artificial Neural Network (ANN)

A Multi-Layer Perceptron (MLP) regressor was used as it is an universal approximators, making it highly suitable for modelling complex, continuous Vapor-Liquid Equilibrium (VLE) phenomena. The network consists of an input layer, multiple hidden layers utilizing activation functions such as ReLU and tanh, and an output layer. The model updates its internal weights and biases through backpropagation, by focusing on minimizing the mean squared error between the actual and the predicted values.

7.2 Support Vector Regression (SVR)

Unlike traditional regression that aims at minimizing the error, SVR attempts to fit the error within a defined threshold (ϵ -tube). To handle the non-linear thermodynamics of the distillation column, the SVR model used the Radial Basis Function (RBF) kernel. The kernel maps the 7 input parameters on a higher dimension and finds a linear regression hyperplane that gives the predicted values.

7.3 Random Forest (RF) Regressor

Random Forest is an ensemble learning method, which operates by constructing a large number of decision trees during training and outputting the average prediction of the individual trees. RF relies on step wise and orthogonal decision boundaries, which is often seen to perform poorly when predicting smooth correlations like non linear thermodynamic relationship.

8 Model Training and Validation Procedure

8.1 Data Splitting

To test the model’s capability to predict completely unseen data, the 5,000-row preprocessed dataset was split into training and testing subsets in 80/20 ratio. The training set (80%) was used exclusively for model fitting and hyperparameter tuning, while the unseen test set (20%) was used to evaluate the model’s performance metrics.

8.2 Hyperparameter Optimization and Cross-Validation

To avoid overfitting and ensure robustness of the model, k-fold cross validation was used (here $K = 5$). This divides the dataset of 5000 into 5 parts (1000 each) and uses permutation by using 4 groups at a time while hiding the 5th group, and at last averages the results. This ensures that the model learns the underlying relationship instead of scoring high due to familiarising itself with the data.

For the Artificial Neural Network (ANN), the following parameter grid was selected:

- **Hidden Layer Sizes:** (64, 32), (100, 50), (128, 64, 32)
- **Activation Functions:** ReLU, Tanh
- **L2 Penalty (α):** 10^{-4} , 10^{-3}
- **Initial Learning Rate:** 10^{-3} , 10^{-2}

For the Support Vector Regression (SVR), standard implementations only support single-target prediction. Therefore, the SVR was wrapped in a `MultiOutputRegressor`. Since SVR scales quadratically with dataset size, a complete grid search was computationally unbearable. Instead, a `RandomizedSearchCV` (8 iterations) was deployed across the following distribution:

- **Regularization (C):** 10, 50, 100, 200
- **Kernel Coefficient (γ):** 'scale', 0.01, 0.1
- **Epsilon-tube (ϵ):** 0.01, 0.05, 0.1

8.3 Performance Metrics

To quantify the predictive accuracy of the trained models against the DWSIM outputs, three primary statistical metrics were evaluated:

1. **Coefficient of Determination (R^2):** Measures the proportion of variance in the dependent variables predictable from the independent variables.
2. **Root Mean Square Error (RMSE):** Measures the standard deviation of the prediction errors (residuals). Because errors are squared before being averaged, RMSE heavily penalizes large outliers, ensuring the models do not suffer from isolated but severe thermodynamic failures.
3. **Mean Absolute Error (MAE):** Provides a linear, interpretable measure of the average prediction error in the original physical units (kW for duty, mole fraction for purity).

9 Performance Comparison of Models

9.1 Optimized Hyperparameters

Following the exhaustive grid search and cross-validation protocols defined in Section 8, the algorithms converged on the following optimal architectures for predicting the distillation thermodynamics:

- **Optimal ANN (MLP):** Two hidden layers (100 neurons, 50 neurons), ReLU activation, L2 regularization ($\alpha = 0.001$), and an initial learning rate of 0.01.
- **Optimal SVR:** Regularization parameter ($C = 50$), epsilon-tube ($\epsilon = 0.05$), and a dynamically scaled RBF kernel coefficient ($\gamma = \text{'scale'}$).

9.2 Evaluation on Unseen Test Data

The fully trained models were evaluated against the withheld 20% test dataset to quantify their real-world generalization capability. The final Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) for all four target variables are consolidated in **Table 2**.

Table 2: Performance comparison of surrogate models on the 20% unseen test set

Output Variable	Random Forest			SVR			ANN (MLP)		
	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2	RMSE	MAE
Condenser Duty (kW)	0.9976	32.5215	26.1126	0.9979	30.0636	20.1072	0.9997	11.4602	9.1442
Reboiler Duty (kW)	0.9985	25.7289	20.6203	0.9980	29.7426	19.8783	0.9997	10.8381	8.5193
Distillate (Benzene)	0.8880	0.0312	0.0252	0.9951	0.0065	0.0050	0.9992	0.0026	0.0020
Bottoms (Toluene)	0.8009	0.0338	0.0292	0.9933	0.0062	0.0047	0.9989	0.0025	0.0018

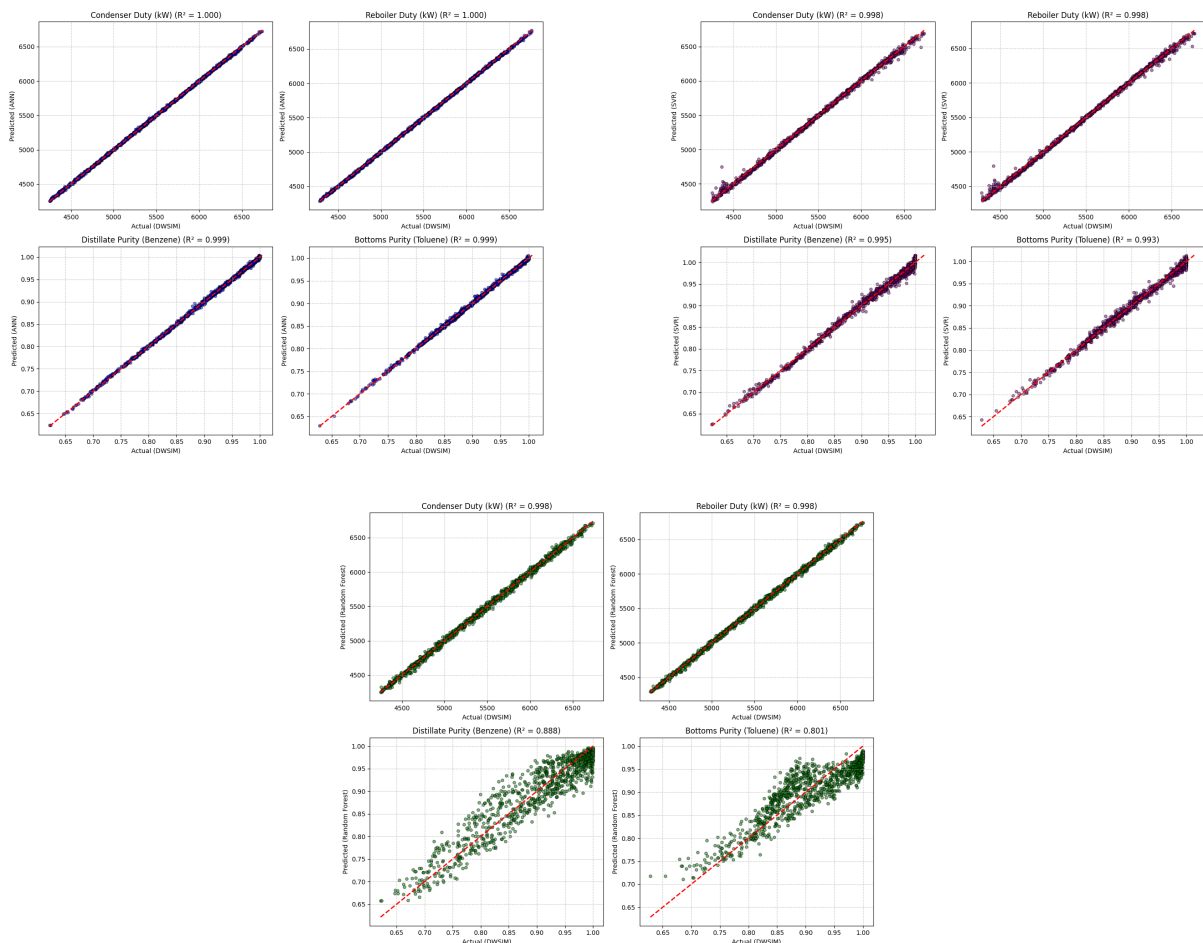


Figure 2: Parity plots comparing the DWSIM ground truth versus predicted values for the ANN, SVR, and RF models on the unseen test set.

10 Discussion on Physical Consistency and Robustness

The stark contrast in performance between the models highlights the critical importance of matching machine learning architectures to the underlying physics of the dataset.

The Random Forest (RF) model demonstrated the poorest performance, particularly in predicting product purities. This was expected, as RF generates orthogonal decision boundaries while VLE curves are continuous as dictated by the Peng-Robinson EoS; they are non-linear and show exponential behaviour. Thus, the model tends to overfit in order to replicate the said smooth relation.

Conversely, the Artificial Neural Network (ANN) and Support Vector Regression (SVR) models exhibited high robustness and physical consistency. The continuous, differentiable nature of the ANN's activation functions and the SVR's Radial Basis Function (RBF) kernel allowed them to smoothly interpolate the 7-dimensional VLE space. Thus, the predictions could accurately identify the cubic roots of the PR EoS without violating any thermodynamic relationships.

11 Final Conclusion

This study successfully developed a fully automated Python-DWSIM COM-bridge to generate a 5,000-point thermodynamic dataset for a Benzene-Toluene distillation column.

Based on the evaluation metrics on unseen test data, the Artificial Neural Network (MLP)

is conclusively identified as the superior surrogate model. By achieving a Coefficient of Determination (R^2) greater than 0.999 and near-zero Mean Absolute Errors across all energy duties and component purities, the ANN successfully mapped the complex Peng-Robinson thermodynamics.

Thus, this study can be concluded by stating that under these specific range of input parameters the ANN model can predict the thermodynamic relations taking less time in the meanwhile and without sacrificing the accuracy of calculation.